WEEKLY TASK BRIEFS BY GROUP

# Digital Forensics Internship

*Weeks 2–8 · description, tools, expected outcome, presentation & report*

**Every listed tool is open-source or free · freeware is marked '(free)'**

## How to use these briefs

Each group has its own set of weekly briefs below (Weeks 2–8). Every brief gives a short description of the task, the open-source or free tools to use, the expected outcome, and the two deliverables. Weeks shaded in a warm tint are research-report weeks.

### Standard deliverables — every week, every group

**Presentation.** A short talk to the cohort at the end of the week — about 3–5 minutes, 5–10 slides. The brief says what to show.

**Report.** A short-written report — about 2–4 pages — in PDF or Word format. The brief says what to document. Research weeks instead produce the research report described in the guidelines at the end.

**Tool policy.** Use only the open-source or free tools named in the briefs (or equivalents agreed with your mentor). Items marked '(free)' are free of charge but not open source.

# Group 1 *— research report in Week 2*

## Week 2 · Research report

**Description:** Cloud forensics — acquiring and validating evidence from cloud services (IaaS/SaaS).

**Tools:** Google Scholar ('Since 2020' filter) & open-access repositories (arXiv, DOAJ); Zotero (reference manager); LibreOffice Writer.

**Expected outcome:** A concise, well-cited review of current (2020-onward) work on the topic, tied back to open-source practice and the Pakistani setting.

**Presentation:** present the state of the art, the main open-source tools involved, and two or three unsolved problems.

**Report:** the 1,200–2,000-word research report; every cited paper dated 2020 or later; IEEE or APA references (see guidelines).

## Week 3 · File-System Forensics & Data Recovery — Alternative tool & comparison

**Description:** Do the same recovery via the Sleuth Kit CLI and compare with Autopsy.

**Tools:** The Sleuth Kit (fls, icat)

**Expected outcome:** A clear comparison showing where each tool is stronger, with matching or explained results.

**Presentation:** show a side-by-side comparison table and give a recommendation.

**Report:** tabulate the comparison and justify the recommendation.

## Week 4 · Windows Artefacts & Registry Forensics — Acquisition / collection

**Description:** Extract the registry hives and event logs from the image safely.

**Tools:** Autopsy / The Sleuth Kit (extract hives & EVTX)

**Expected outcome:** A forensically sound collection with a complete, verifiable log and matching hashes.

**Presentation:** show the acquisition steps and the integrity proof (before/after hashes).

**Report:** a full acquisition log: source, method, tool version, hashes and times.

## Week 5 · Memory (RAM) Forensics — Core analysis

**Description:** Enumerate processes and identify any parent/child anomalies.

**Tools:** Volatility 3 (pslist, pstree)

**Expected outcome:** The week's key artefacts correctly extracted and interpreted.

**Presentation:** show the artefacts found and what they mean for the case.

**Report:** the artefacts, their location and interpretation, with limitations stated.

## Week 6 · Network Forensics — Recovery / extraction

**Description:** Extract transferred files, images or credentials from the PCAP.

**Tools:** NetworkMiner (free)

**Expected outcome:** Recovered items catalogued, with method and confidence noted.

**Presentation:** show a before/after of what was recovered and how.

**Report:** a catalogue of recovered items with method and confidence.

## Week 7 · Mobile Forensics & Malware Triage — Timeline & correlation

**Description:** Build a device-usage timeline and correlate it with the case narrative.
**Tools:** ALEAPP timeline, Plaso
**Expected outcome:** A readable timeline that links this week's evidence to the wider case.
**Presentation:** show the timeline visual and the story it tells.
**Report:** the timeline and the reasoning behind the reconstructed sequence.

## Week 8 · Anti-Forensics, Reporting & Project Delivery — Anti-forensics angle

**Description:** Write the 'anti-forensic activity found / ruled out' section.
**Tools:** MFTECmd, ExifTool, Autopsy (document tampering)
**Expected outcome:** Evidence of what concealment leaves behind and a reliable way to detect it.
**Presentation:** show the concealment attempted and the detection method that caught it.
**Report:** the technique, the residual traces and the detection steps.

# Group 2 *— research report in Week 2*

## Week 2 · Research report

**Description:** Forensic implications of full-disk and application-level encryption.

**Tools:** Google Scholar ('Since 2020' filter) & open-access repositories (arXiv, DOAJ); Zotero (reference manager); LibreOffice Writer.

**Expected outcome:** A concise, well-cited review of current (2020-onward) work on the topic, tied back to open-source practice and the Pakistani setting.

**Presentation:** present the state of the art, the main open-source tools involved, and two or three unsolved problems.

**Report:** the 1,200–2,000-word research report; every cited paper dated 2020 or later; IEEE or APA references (see guidelines).

## Week 3 · File-System Forensics & Data Recovery — Acquisition / collection

**Description:** Mount the image read-only across FAT/NTFS/ext and document the options used.

**Tools:** mount (read-only), The Sleuth Kit

**Expected outcome:** A forensically sound collection with a complete, verifiable log and matching hashes.

**Presentation:** show the acquisition steps and the integrity proof (before/after hashes).

**Report:** a full acquisition log: source, method, tool version, hashes and times.

## Week 4 · Windows Artefacts & Registry Forensics — Core analysis

**Description:** Interpret execution artefacts (Prefetch, LNK, Jump Lists).

**Tools:** PECmd, LECmd, JLECmd (Eric Zimmerman tools, free)

**Expected outcome:** The week's key artefacts correctly extracted and interpreted.

**Presentation:** show the artefacts found and what they mean for the case.

**Report:** the artefacts, their location and interpretation, with limitations stated.

## Week 5 · Memory (RAM) Forensics — Recovery / extraction

**Description:** Extract network connections, command history and cached credentials.

**Tools:** Volatility 3 (netscan, cmdline, hashdump)

**Expected outcome:** Recovered items catalogued, with method and confidence noted.

**Presentation:** show a before/after of what was recovered and how.

**Report:** a catalogue of recovered items with method and confidence.

## Week 6 · Network Forensics — Timeline & correlation

**Description:** Correlate network events with host artefacts from earlier weeks.

**Tools:** Zeek logs, tshark, Arkime

**Expected outcome:** A readable timeline that links this week's evidence to the wider case.

**Presentation:** show the timeline visual and the story it tells.

**Report:** the timeline and the reasoning behind the reconstructed sequence.

## Week 7 · Mobile Forensics & Malware Triage — Anti-forensics angle

**Description:** Examine app-data obfuscation and detonate a sample safely in a sandbox.

**Tools:** MobSF, CAPE / Cuckoo sandbox

**Expected outcome:** Evidence of what concealment leaves behind and a reliable way to detect it.

**Presentation:** show the concealment attempted and the detection method that caught it.

**Report:** the technique, the residual traces and the detection steps.

## Week 8 · Anti-Forensics, Reporting & Project Delivery — Integrity & validation (QA)

**Description:** Perform final QA on the whole report and confirm the chain of custody is complete.

**Tools:** md5sum, sha256sum, hashdeep

**Expected outcome:** Independent confirmation (or refutation) of another group's results, backed by hashes.

**Presentation:** show your verification method and whether the results matched.

**Report:** your independent results and a pass/fail against the original.

# Group 3 *— research report in Week 3*

## Week 2 · Evidence Acquisition & Imaging — Acquisition / collection

**Description:** Run a dead-box acquisition end to end and produce a complete acquisition log.

**Tools:** dc3dd, FTK Imager (free)

**Expected outcome:** A forensically sound collection with a complete, verifiable log and matching hashes.

**Presentation:** show the acquisition steps and the integrity proof (before/after hashes).

**Report:** a full acquisition log: source, method, tool version, hashes and times.

## Week 3 · Research report

**Description:** SSD forensics — how TRIM and wear-levelling affect data recoverability.

**Tools:** Google Scholar ('Since 2020' filter) & open-access repositories (arXiv, DOAJ); Zotero (reference manager); LibreOffice Writer.

**Expected outcome:** A concise, well-cited review of current (2020-onward) work on the topic, tied back to open-source practice and the Pakistani setting.

**Presentation:** present the state of the art, the main open-source tools involved, and two or three unsolved problems.

**Report:** the 1,200–2,000-word research report; every cited paper dated 2020 or later; IEEE or APA references (see guidelines).

## Week 4 · Windows Artefacts & Registry Forensics — Recovery / extraction

**Description:** Recover browser history and downloads.

**Tools:** Hindsight, Autopsy

**Expected outcome:** Recovered items catalogued, with method and confidence noted.

**Presentation:** show a before/after of what was recovered and how.

**Report:** a catalogue of recovered items with method and confidence.

## Week 5 · Memory (RAM) Forensics — Timeline & correlation

**Description:** Correlate in-memory findings with the disk artefacts from Weeks 3–4.

**Tools:** Volatility 3 (timeliner), Plaso

**Expected outcome:** A readable timeline that links this week's evidence to the wider case.

**Presentation:** show the timeline visual and the story it tells.

**Report:** the timeline and the reasoning behind the reconstructed sequence.

## Week 6 · Network Forensics — Anti-forensics angle

**Description:** Examine encrypted/tunnelled traffic and note what can still be inferred.

**Tools:** Wireshark (TLS analysis), tshark

**Expected outcome:** Evidence of what concealment leaves behind and a reliable way to detect it.

**Presentation:** show the concealment attempted and the detection method that caught it.

**Report:** the technique, the residual traces and the detection steps.

## Week 7 · Mobile Forensics & Malware Triage — Integrity & validation (QA)

**Description:** Validate another group's extraction/triage steps for reproducibility.

**Tools:** md5sum, sha256sum, hashdeep

**Expected outcome:** Independent confirmation (or refutation) of another group's results, backed by hashes.

**Presentation:** show your verification method and whether the results matched.

**Report:** your independent results and a pass/fail against the original.

## Week 8 · Anti-Forensics, Reporting & Project Delivery — SOP & handbook

**Description:** Assemble and finalise the full Lab & SOP Handbook from all the weekly SOPs.

**Tools:** LibreOffice Writer / a Markdown editor

**Expected outcome:** A concise, reproducible SOP another examiner could follow unaided.

**Presentation:** walk through the SOP's key steps and the pitfalls to avoid.

**Report:** the finished SOP, ready to drop into the project handbook.

# Group 4 *— research report in Week 3*

## Week 2 · Evidence Acquisition & Imaging — Core analysis

**Description:** Confirm the partition layout and preview the image strictly read-only.

**Tools:** mmls (The Sleuth Kit), read-only loop mount

**Expected outcome:** The week's key artefacts correctly extracted and interpreted.

**Presentation:** show the artefacts found and what they mean for the case.

**Report:** the artefacts, their location and interpretation, with limitations stated.

## Week 3 · Research report

**Description:** Recent advances in file carving and fragmented-file recovery.

**Tools:** Google Scholar ('Since 2020' filter) & open-access repositories (arXiv, DOAJ); Zotero (reference manager); LibreOffice Writer.

**Expected outcome:** A concise, well-cited review of current (2020-onward) work on the topic, tied back to open-source practice and the Pakistani setting.

**Presentation:** present the state of the art, the main open-source tools involved, and two or three unsolved problems.

**Report:** the 1,200–2,000-word research report; every cited paper dated 2020 or later; IEEE or APA references (see guidelines).

## Week 4 · Windows Artefacts & Registry Forensics — Timeline & correlation

**Description:** Merge artefacts into one user-activity timeline and correlate with Week 3.

**Tools:** Plaso / log2timeline, Timeline Explorer (free)

**Expected outcome:** A readable timeline that links this week's evidence to the wider case.

**Presentation:** show the timeline visual and the story it tells.

**Report:** the timeline and the reasoning behind the reconstructed sequence.

## Week 5 · Memory (RAM) Forensics — Anti-forensics angle

**Description:** Examine process-injection/hollowing indicators and document detection.

**Tools:** Volatility 3 (malfind, ldrmodules)

**Expected outcome:** Evidence of what concealment leaves behind and a reliable way to detect it.

**Presentation:** show the concealment attempted and the detection method that caught it.

**Report:** the technique, the residual traces and the detection steps.

## Week 6 · Network Forensics — Integrity & validation (QA)

**Description:** Validate another group's session reconstruction against the raw PCAP.

**Tools:** md5sum, sha256sum, hashdeep

**Expected outcome:** Independent confirmation (or refutation) of another group's results, backed by hashes.

**Presentation:** show your verification method and whether the results matched.

**Report:** your independent results and a pass/fail against the original.

## Week 7 · Mobile Forensics & Malware Triage — SOP & handbook

**Description:** Draft the Mobile & Malware Triage SOP.

**Tools:** LibreOffice Writer / a Markdown editor

**Expected outcome:** A concise, reproducible SOP another examiner could follow unaided.
**Presentation:** walk through the SOP's key steps and the pitfalls to avoid.
**Report:** the finished SOP, ready to drop into the project handbook.

## Week 8 · Anti-Forensics, Reporting & Project Delivery — Legal & admissibility

**Description:** Write the scope/authorisation/admissibility section citing PECA 2016 and the QSO.
**Tools:** No software — PECA 2016 & Qanun-e-Shahadat Order texts; reported judgments
**Expected outcome:** A clear mapping from the technical steps to admissibility requirements.
**Presentation:** present the legal mapping and the top admissibility risks.
**Report:** the mapping to PECA 2016 / the QSO and the risks to admissibility.

# Group 5 *— research report in Week 4*

## Week 2 · Evidence Acquisition & Imaging — Recovery / extraction

**Description:** Capture the unallocated/slack area of the image and record sizes for later carving.
**Tools:** The Sleuth Kit (blkls), dd
**Expected outcome:** Recovered items catalogued, with method and confidence noted.
**Presentation:** show a before/after of what was recovered and how.
**Report:** a catalogue of recovered items with method and confidence.

## Week 3 · File-System Forensics & Data Recovery — Timeline & correlation

**Description:** Build a super-timeline and pick out a plausible sequence of user activity.
**Tools:** Plaso / log2timeline, Timesketch
**Expected outcome:** A readable timeline that links this week's evidence to the wider case.
**Presentation:** show the timeline visual and the story it tells.
**Report:** the timeline and the reasoning behind the reconstructed sequence.

## Week 4 · Research report

**Description:** Reconstructing user activity from Windows artefacts with current open-source tooling.
**Tools:** Google Scholar ('Since 2020' filter) & open-access repositories (arXiv, DOAJ); Zotero (reference manager); LibreOffice Writer.
**Expected outcome:** A concise, well-cited review of current (2020-onward) work on the topic, tied back to open-source practice and the Pakistani setting.
**Presentation:** present the state of the art, the main open-source tools involved, and two or three unsolved problems.
**Report:** the 1,200–2,000-word research report; every cited paper dated 2020 or later; IEEE or APA references (see guidelines).

## Week 5 · Memory (RAM) Forensics — Integrity & validation (QA)

**Description:** Reproduce another group's Volatility results and confirm consistency.
**Tools:** md5sum, sha256sum, hashdeep
**Expected outcome:** Independent confirmation (or refutation) of another group's results, backed by hashes.
**Presentation:** show your verification method and whether the results matched.
**Report:** your independent results and a pass/fail against the original.

## Week 6 · Network Forensics — SOP & handbook

**Description:** Draft the Network Forensics SOP.
**Tools:** LibreOffice Writer / a Markdown editor
**Expected outcome:** A concise, reproducible SOP another examiner could follow unaided.
**Presentation:** walk through the SOP's key steps and the pitfalls to avoid.
**Report:** the finished SOP, ready to drop into the project handbook.

## Week 7 · Mobile Forensics & Malware Triage — Legal & admissibility

**Description:** Discuss consent and scope for phone searches and lawful handling of personal data.
**Tools:** No software — PECA 2016 & Qanun-e-Shahadat Order texts; reported judgments

**Expected outcome:** A clear mapping from the technical steps to admissibility requirements.
**Presentation:** present the legal mapping and the top admissibility risks.
**Report:** the mapping to PECA 2016 / the QSO and the risks to admissibility.

## Week 8 · Anti-Forensics, Reporting & Project Delivery — Scenario / dataset builder

**Description:** Prepare the exhibit index and appendices for the report.
**Tools:** Autopsy report module, LibreOffice Writer (exhibit index & appendices)
**Expected outcome:** A realistic, documented practice dataset with an answer key/QA checklist.
**Presentation:** explain the scenario design and what it is meant to test.
**Report:** the dataset description, how it was built, and the answer key.

# Group 6 *— research report in Week 4*

## Week 2 · Evidence Acquisition & Imaging — Timeline & correlation

**Description:** Enter all acquisition metadata (source, method, hashes, times) into the master evidence register.

**Tools:** LibreOffice Calc (evidence register)

**Expected outcome:** A readable timeline that links this week's evidence to the wider case.

**Presentation:** show the timeline visual and the story it tells.

**Report:** the timeline and the reasoning behind the reconstructed sequence.

## Week 3 · File-System Forensics & Data Recovery — Anti-forensics angle

**Description:** Wipe, rename and extension-spoof test files, then show how carving/signatures defeat it.

**Tools:** shred (wipe test), PhotoRec, bulk_extractor

**Expected outcome:** Evidence of what concealment leaves behind and a reliable way to detect it.

**Presentation:** show the concealment attempted and the detection method that caught it.

**Report:** the technique, the residual traces and the detection steps.

## Week 4 · Research report

**Description:** Detecting anti-forensics — timestamp manipulation, wiping and artefact tampering.

**Tools:** Google Scholar ('Since 2020' filter) & open-access repositories (arXiv, DOAJ); Zotero (reference manager); LibreOffice Writer.

**Expected outcome:** A concise, well-cited review of current (2020-onward) work on the topic, tied back to open-source practice and the Pakistani setting.

**Presentation:** present the state of the art, the main open-source tools involved, and two or three unsolved problems.

**Report:** the 1,200–2,000-word research report; every cited paper dated 2020 or later; IEEE or APA references (see guidelines).

## Week 5 · Memory (RAM) Forensics — SOP & handbook

**Description:** Draft the Memory Forensics SOP.

**Tools:** LibreOffice Writer / a Markdown editor

**Expected outcome:** A concise, reproducible SOP another examiner could follow unaided.

**Presentation:** walk through the SOP's key steps and the pitfalls to avoid.

**Report:** the finished SOP, ready to drop into the project handbook.

## Week 6 · Network Forensics — Legal & admissibility

**Description:** Map the work to lawful-interception / provider-record realities and cross-border issues.

**Tools:** No software — PECA 2016 & Qanun-e-Shahadat Order texts; reported judgments

**Expected outcome:** A clear mapping from the technical steps to admissibility requirements.

**Presentation:** present the legal mapping and the top admissibility risks.

**Report:** the mapping to PECA 2016 / the QSO and the risks to admissibility.

## Week 7 · Mobile Forensics & Malware Triage — Scenario / dataset builder

**Description:** Prepare a test Android image/app plus a safe malware-triage sample for others.
**Tools:** Android emulator (AVD), the EICAR test file
**Expected outcome:** A realistic, documented practice dataset with an answer key/QA checklist.
**Presentation:** explain the scenario design and what it is meant to test.
**Report:** the dataset description, how it was built, and the answer key.

## Week 8 · Anti-Forensics, Reporting & Project Delivery — Presentation & cheat-sheet

**Description:** Prepare and lead the final briefing to the mentor.
**Tools:** LibreOffice Impress (plus the week's own outputs)
**Expected outcome:** A shared understanding across all groups and a reusable one-page reference.
**Presentation:** lead the share-out and summarise every group's findings for the week.
**Report:** a one-page cheat-sheet plus a short summary of the week's collective findings.

# Group 7 *— research report in Week 5*

## Week 2 · Evidence Acquisition & Imaging — Anti-forensics angle

**Description:** Attempt to acquire an encrypted/hidden test volume and document what is and isn't obtainable.

**Tools:** VeraCrypt (to build the test volume), Guymager

**Expected outcome:** Evidence of what concealment leaves behind and a reliable way to detect it.

**Presentation:** show the concealment attempted and the detection method that caught it.

**Report:** the technique, the residual traces and the detection steps.

## Week 3 · File-System Forensics & Data Recovery — Integrity & validation (QA)

**Description:** Cross-check another group's recovered-file hashes and note the false-positive rate.

**Tools:** md5sum, sha256sum, hashdeep

**Expected outcome:** Independent confirmation (or refutation) of another group's results, backed by hashes.

**Presentation:** show your verification method and whether the results matched.

**Report:** your independent results and a pass/fail against the original.

## Week 4 · Windows Artefacts & Registry Forensics — SOP & handbook

**Description:** Draft the Windows Artefact SOP.

**Tools:** LibreOffice Writer / a Markdown editor

**Expected outcome:** A concise, reproducible SOP another examiner could follow unaided.

**Presentation:** walk through the SOP's key steps and the pitfalls to avoid.

**Report:** the finished SOP, ready to drop into the project handbook.

## Week 5 · Research report

**Description:** Memory forensics for detecting fileless and in-memory malware.

**Tools:** Google Scholar ('Since 2020' filter) & open-access repositories (arXiv, DOAJ); Zotero (reference manager); LibreOffice Writer.

**Expected outcome:** A concise, well-cited review of current (2020-onward) work on the topic, tied back to open-source practice and the Pakistani setting.

**Presentation:** present the state of the art, the main open-source tools involved, and two or three unsolved problems.

**Report:** the 1,200–2,000-word research report; every cited paper dated 2020 or later; IEEE or APA references (see guidelines).

## Week 6 · Network Forensics — Scenario / dataset builder

**Description:** Prepare a PCAP scenario (e.g. simulated phishing or file exfiltration) for others.

**Tools:** tcpreplay, a lab VM, a self-built phishing page (kept offline)

**Expected outcome:** A realistic, documented practice dataset with an answer key/QA checklist.

**Presentation:** explain the scenario design and what it is meant to test.

**Report:** the dataset description, how it was built, and the answer key.

## Week 7 · Mobile Forensics & Malware Triage — Presentation & cheat-sheet

**Description:** Run the weekly share-out and produce a triage cheat-sheet.
**Tools:** LibreOffice Impress (plus the week's own outputs)
**Expected outcome:** A shared understanding across all groups and a reusable one-page reference.
**Presentation:** lead the share-out and summarise every group's findings for the week.
**Report:** a one-page cheat-sheet plus a short summary of the week's collective findings.

## Week 8 · Anti-Forensics, Reporting & Project Delivery — Primary tool lead

**Description:** Run steganography and embedded-file checks across the case evidence.
**Tools:** steghide, zsteg, binwalk
**Expected outcome:** A repeatable procedure for the week's main tool, with a verified correct result.
**Presentation:** walk the cohort through the tool workflow (live or screenshots) and show one verified result.
**Report:** document install, commands and a verified example, with screenshots.

# Group 8 *— research report in Week 5*

## Week 2 · Evidence Acquisition & Imaging — Integrity & validation (QA)

**Description:** Independently re-hash another group's image and confirm the match.
**Tools:** md5sum, sha256sum, hashdeep
**Expected outcome:** Independent confirmation (or refutation) of another group's results, backed by hashes.
**Presentation:** show your verification method and whether the results matched.
**Report:** your independent results and a pass/fail against the original.

## Week 3 · File-System Forensics & Data Recovery — SOP & handbook

**Description:** Draft the File-System Analysis & Recovery SOP.
**Tools:** LibreOffice Writer / a Markdown editor
**Expected outcome:** A concise, reproducible SOP another examiner could follow unaided.
**Presentation:** walk through the SOP's key steps and the pitfalls to avoid.
**Report:** the finished SOP, ready to drop into the project handbook.

## Week 4 · Windows Artefacts & Registry Forensics — Legal & admissibility

**Description:** Explain 'activity is not identity' and how artefacts are presented in court.
**Tools:** No software — PECA 2016 & Qanun-e-Shahadat Order texts; reported judgments
**Expected outcome:** A clear mapping from the technical steps to admissibility requirements.
**Presentation:** present the legal mapping and the top admissibility risks.
**Report:** the mapping to PECA 2016 / the QSO and the risks to admissibility.

## Week 5 · Research report

**Description:** Machine-learning approaches to digital-forensic triage and malware detection.
**Tools:** Google Scholar ('Since 2020' filter) & open-access repositories (arXiv, DOAJ); Zotero (reference manager); LibreOffice Writer.
**Expected outcome:** A concise, well-cited review of current (2020-onward) work on the topic, tied back to open-source practice and the Pakistani setting.
**Presentation:** present the state of the art, the main open-source tools involved, and two or three unsolved problems.
**Report:** the 1,200–2,000-word research report; every cited paper dated 2020 or later; IEEE or APA references (see guidelines).

## Week 6 · Network Forensics — Presentation & cheat-sheet

**Description:** Run the weekly share-out and produce a Wireshark filter cheat-sheet.
**Tools:** LibreOffice Impress (plus the week's own outputs)
**Expected outcome:** A shared understanding across all groups and a reusable one-page reference.
**Presentation:** lead the share-out and summarise every group's findings for the week.
**Report:** a one-page cheat-sheet plus a short summary of the week's collective findings.

## Week 7 · Mobile Forensics & Malware Triage — Primary tool lead

**Description:** Extract Android logical data via ADB and parse it with ALEAPP.
**Tools:** ADB + ALEAPP

**Expected outcome:** A repeatable procedure for the week's main tool, with a verified correct result.

**Presentation:** walk the cohort through the tool workflow (live or screenshots) and show one verified result.

**Report:** document install, commands and a verified example, with screenshots.

### Week 8 · Anti-Forensics, Reporting & Project Delivery — Alternative tool & comparison

**Description:** Cross-check timestamps for tampering and compare the tools' verdicts.

**Tools:** stegdetect, ExifTool (timestamp cross-check)

**Expected outcome:** A clear comparison showing where each tool is stronger, with matching or explained results.

**Presentation:** show a side-by-side comparison table and give a recommendation.

**Report:** tabulate the comparison and justify the recommendation.

# Group 9 *— research report in Week 6*

## Week 2 · Evidence Acquisition & Imaging — SOP & handbook

**Description:** Draft the Acquisition & Integrity SOP for the project handbook.
**Tools:** LibreOffice Writer / a Markdown editor
**Expected outcome:** A concise, reproducible SOP another examiner could follow unaided.
**Presentation:** walk through the SOP's key steps and the pitfalls to avoid.
**Report:** the finished SOP, ready to drop into the project handbook.

## Week 3 · File-System Forensics & Data Recovery — Legal & admissibility

**Description:** Note how deleted/recovered data is characterised evidentially and any authenticity concerns.
**Tools:** No software — PECA 2016 & Qanun-e-Shahadat Order texts; reported judgments
**Expected outcome:** A clear mapping from the technical steps to admissibility requirements.
**Presentation:** present the legal mapping and the top admissibility risks.
**Report:** the mapping to PECA 2016 / the QSO and the risks to admissibility.

## Week 4 · Windows Artefacts & Registry Forensics — Scenario / dataset builder

**Description:** Prepare a Windows artefact set (USB history, browsing, program runs) for others.
**Tools:** a Windows test VM, spare USB sticks, a web browser (to generate artefacts)
**Expected outcome:** A realistic, documented practice dataset with an answer key/QA checklist.
**Presentation:** explain the scenario design and what it is meant to test.
**Report:** the dataset description, how it was built, and the answer key.

## Week 5 · Memory (RAM) Forensics — Presentation & cheat-sheet

**Description:** Run the weekly share-out and produce a Volatility plugin cheat-sheet.
**Tools:** LibreOffice Impress (plus the week's own outputs)
**Expected outcome:** A shared understanding across all groups and a reusable one-page reference.
**Presentation:** lead the share-out and summarise every group's findings for the week.
**Report:** a one-page cheat-sheet plus a short summary of the week's collective findings.

## Week 6 · Research report

**Description:** Network forensics of encrypted traffic (TLS 1.3, QUIC, DNS-over-HTTPS).
**Tools:** Google Scholar ('Since 2020' filter) & open-access repositories (arXiv, DOAJ); Zotero (reference manager); LibreOffice Writer.
**Expected outcome:** A concise, well-cited review of current (2020-onward) work on the topic, tied back to open-source practice and the Pakistani setting.
**Presentation:** present the state of the art, the main open-source tools involved, and two or three unsolved problems.
**Report:** the 1,200–2,000-word research report; every cited paper dated 2020 or later; IEEE or APA references (see guidelines).

## Week 7 · Mobile Forensics & Malware Triage — Alternative tool & comparison

**Description:** Parse the same data with Andriller and compare coverage.

**Tools:** Andriller (Community Edition)

**Expected outcome:** A clear comparison showing where each tool is stronger, with matching or explained results.

**Presentation:** show a side-by-side comparison table and give a recommendation.

**Report:** tabulate the comparison and justify the recommendation.

## Week 8 · Anti-Forensics, Reporting & Project Delivery — Acquisition / collection

**Description:** Re-assemble and re-verify all case exhibits and their hashes for the final report.

**Tools:** hashdeep (re-verify exhibits)

**Expected outcome:** A forensically sound collection with a complete, verifiable log and matching hashes.

**Presentation:** show the acquisition steps and the integrity proof (before/after hashes).

**Report:** a full acquisition log: source, method, tool version, hashes and times.

# Group 10 *— research report in Week 6*

## Week 2 · Evidence Acquisition & Imaging — Legal & admissibility

**Description:** Map imaging and hashing to chain-of-custody duties under the Qanun-e-Shahadat Order.

**Tools:** No software — PECA 2016 & Qanun-e-Shahadat Order texts; reported judgments

**Expected outcome:** A clear mapping from the technical steps to admissibility requirements.

**Presentation:** present the legal mapping and the top admissibility risks.

**Report:** the mapping to PECA 2016 / the QSO and the risks to admissibility.

## Week 3 · File-System Forensics & Data Recovery — Scenario / dataset builder

**Description:** Seed an image with deleted, hidden and renamed files for other groups to recover.

**Tools:** dd, mkfs, shred, mv (to seed the image)

**Expected outcome:** A realistic, documented practice dataset with an answer key/QA checklist.

**Presentation:** explain the scenario design and what it is meant to test.

**Report:** the dataset description, how it was built, and the answer key.

## Week 4 · Windows Artefacts & Registry Forensics — Presentation & cheat-sheet

**Description:** Run the weekly share-out and produce a Windows-artefacts cheat-sheet.

**Tools:** LibreOffice Impress (plus the week's own outputs)

**Expected outcome:** A shared understanding across all groups and a reusable one-page reference.

**Presentation:** lead the share-out and summarise every group's findings for the week.

**Report:** a one-page cheat-sheet plus a short summary of the week's collective findings.

## Week 5 · Memory (RAM) Forensics — Primary tool lead

**Description:** Analyse a memory image in Volatility 3 and document the plugin workflow.

**Tools:** Volatility 3

**Expected outcome:** A repeatable procedure for the week's main tool, with a verified correct result.

**Presentation:** walk the cohort through the tool workflow (live or screenshots) and show one verified result.

**Report:** document install, commands and a verified example, with screenshots.

## Week 6 · Research report

**Description:** Investigating phishing and online financial fraud — with a Pakistan focus.

**Tools:** Google Scholar ('Since 2020' filter) & open-access repositories (arXiv, DOAJ); Zotero (reference manager); LibreOffice Writer.

**Expected outcome:** A concise, well-cited review of current (2020-onward) work on the topic, tied back to open-source practice and the Pakistani setting.

**Presentation:** present the state of the art, the main open-source tools involved, and two or three unsolved problems.

**Report:** the 1,200–2,000-word research report; every cited paper dated 2020 or later; IEEE or APA references (see guidelines).

## Week 7 · Mobile Forensics & Malware Triage — Acquisition / collection

**Description:** Document logical vs file-system vs physical acquisition trade-offs on a test device.
**Tools:** ADB, libimobiledevice (iOS)
**Expected outcome:** A forensically sound collection with a complete, verifiable log and matching hashes.
**Presentation:** show the acquisition steps and the integrity proof (before/after hashes).
**Report:** a full acquisition log: source, method, tool version, hashes and times.

### Week 8 · Anti-Forensics, Reporting & Project Delivery — Core analysis

**Description:** Compile the consolidated findings section of the final report.
**Tools:** Autopsy (report module)
**Expected outcome:** The week's key artefacts correctly extracted and interpreted.
**Presentation:** show the artefacts found and what they mean for the case.
**Report:** the artefacts, their location and interpretation, with limitations stated.

# Group 11 *— research report in Week 7*

## Week 2 · Evidence Acquisition & Imaging — Scenario / dataset builder

**Description:** Prepare the seeded test media the other groups will acquire this week.
**Tools:** dd, mkfs, a hex editor (wxHexEditor)
**Expected outcome:** A realistic, documented practice dataset with an answer key/QA checklist.
**Presentation:** explain the scenario design and what it is meant to test.
**Report:** the dataset description, how it was built, and the answer key.

## Week 3 · File-System Forensics & Data Recovery — Presentation & cheat-sheet

**Description:** Run the weekly share-out on recovery and produce a carving + timeline cheat-sheet.
**Tools:** LibreOffice Impress (plus the week's own outputs)
**Expected outcome:** A shared understanding across all groups and a reusable one-page reference.
**Presentation:** lead the share-out and summarise every group's findings for the week.
**Report:** a one-page cheat-sheet plus a short summary of the week's collective findings.

## Week 4 · Windows Artefacts & Registry Forensics — Primary tool lead

**Description:** Parse the registry hives with RegRipper and document the key findings.
**Tools:** RegRipper
**Expected outcome:** A repeatable procedure for the week's main tool, with a verified correct result.
**Presentation:** walk the cohort through the tool workflow (live or screenshots) and show one verified result.
**Report:** document install, commands and a verified example, with screenshots.

## Week 5 · Memory (RAM) Forensics — Alternative tool & comparison

**Description:** Mount the same image with MemProcFS and compare browsing versus plugins.
**Tools:** MemProcFS
**Expected outcome:** A clear comparison showing where each tool is stronger, with matching or explained results.
**Presentation:** show a side-by-side comparison table and give a recommendation.
**Report:** tabulate the comparison and justify the recommendation.

## Week 6 · Network Forensics — Acquisition / collection

**Description:** Capture lab traffic with tcpdump and document capture filters and privacy handling.
**Tools:** tcpdump
**Expected outcome:** A forensically sound collection with a complete, verifiable log and matching hashes.
**Presentation:** show the acquisition steps and the integrity proof (before/after hashes).
**Report:** a full acquisition log: source, method, tool version, hashes and times.

## Week 7 · Research report

**Description:** Mobile forensics on modern, hardened Android and iOS devices.
**Tools:** Google Scholar ('Since 2020' filter) & open-access repositories (arXiv, DOAJ); Zotero (reference manager); LibreOffice Writer.

**Expected outcome:** A concise, well-cited review of current (2020-onward) work on the topic, tied back to open-source practice and the Pakistani setting.

**Presentation:** present the state of the art, the main open-source tools involved, and two or three unsolved problems.

**Report:** the 1,200–2,000-word research report; every cited paper dated 2020 or later; IEEE or APA references (see guidelines).

## Week 8 · Anti-Forensics, Reporting & Project Delivery — Recovery / extraction

**Description:** Do a final sweep for concealed data across the case and document the results.

**Tools:** binwalk, foremost (final concealed-data sweep)

**Expected outcome:** Recovered items catalogued, with method and confidence noted.

**Presentation:** show a before/after of what was recovered and how.

**Report:** a catalogue of recovered items with method and confidence.

# Group 12 *— research report in Week 7*

## Week 2 · Evidence Acquisition & Imaging — Presentation & cheat-sheet

**Description:** Run the weekly share-out on acquisition and produce a 'safe imaging' cheat-sheet.
**Tools:** LibreOffice Impress (plus the week's own outputs)
**Expected outcome:** A shared understanding across all groups and a reusable one-page reference.
**Presentation:** lead the share-out and summarise every group's findings for the week.
**Report:** a one-page cheat-sheet plus a short summary of the week's collective findings.

## Week 3 · File-System Forensics & Data Recovery — Primary tool lead

**Description:** Analyse the image in Autopsy and document the ingest and keyword-search workflow.
**Tools:** Autopsy
**Expected outcome:** A repeatable procedure for the week's main tool, with a verified correct result.
**Presentation:** walk the cohort through the tool workflow (live or screenshots) and show one verified result.
**Report:** document install, commands and a verified example, with screenshots.

## Week 4 · Windows Artefacts & Registry Forensics — Alternative tool & comparison

**Description:** Re-examine the same hives with Registry Explorer and compare interpretation depth.
**Tools:** Registry Explorer (Eric Zimmerman tools, free)
**Expected outcome:** A clear comparison showing where each tool is stronger, with matching or explained results.
**Presentation:** show a side-by-side comparison table and give a recommendation.
**Report:** tabulate the comparison and justify the recommendation.

## Week 5 · Memory (RAM) Forensics — Acquisition / collection

**Description:** Capture RAM from a test VM with order-of-volatility notes.
**Tools:** winpmem, DumpIt (free), LiME
**Expected outcome:** A forensically sound collection with a complete, verifiable log and matching hashes.
**Presentation:** show the acquisition steps and the integrity proof (before/after hashes).
**Report:** a full acquisition log: source, method, tool version, hashes and times.

## Week 6 · Network Forensics — Core analysis

**Description:** Reconstruct one session and identify the protocols and endpoints involved.
**Tools:** Wireshark (Follow TCP Stream), tshark
**Expected outcome:** The week's key artefacts correctly extracted and interpreted.
**Presentation:** show the artefacts found and what they mean for the case.
**Report:** the artefacts, their location and interpretation, with limitations stated.

## Week 7 · Research report

**Description:** IoT and smart-device forensics — sources of evidence and extraction methods.

**Tools:** Google Scholar ('Since 2020' filter) & open-access repositories (arXiv, DOAJ); Zotero (reference manager); LibreOffice Writer.

**Expected outcome:** A concise, well-cited review of current (2020-onward) work on the topic, tied back to open-source practice and the Pakistani setting.

**Presentation:** present the state of the art, the main open-source tools involved, and two or three unsolved problems.

**Report:** the 1,200–2,000-word research report; every cited paper dated 2020 or later; IEEE or APA references (see guidelines).

## Week 8 · Anti-Forensics, Reporting & Project Delivery — Timeline & correlation

**Description:** Produce the master case timeline integrating every evidence type.

**Tools:** Plaso, Timesketch

**Expected outcome:** A readable timeline that links this week's evidence to the wider case.

**Presentation:** show the timeline visual and the story it tells.

**Report:** the timeline and the reasoning behind the reconstructed sequence.

# Research report guidelines (2020-onward sources)

Six groups' weeks are research weeks (two groups in each of Weeks 2–7). Each research report is short — about 1,200–2,000 words / 3–4 pages — and surveys current work on the assigned topic.

## The source rule

Every cited research paper must be published in 2020 or later; check the year before citing. Use at least five sources, mostly peer reviewed. If a key idea is older, cite a 2020-or-later paper that reviews or builds on it. Mark preprints (e.g. arXiv) as non-peer reviewed.

## Where to look

Forensic Science International: Digital Investigation (DFRWS) and DFRWS proceedings; IEEE (TIFS, Access), the ACM Digital Library and SpringerLink; Google Scholar with the 'Since 2020' filter; prefer open-access articles so the whole group can read the full text.

## Structure & integrity

Introduction and scope; background; current techniques and tools (favour open-source); challenges and limitations; relevance to Pakistani practice (PECA 2016 / the Qanun-e-Shahadat Order); conclusion and references in a consistent style. Write in your own words, cite every source, and state the licence of any tool discussed.

## Topics by group

| Group | Week | Topic |
|---|---|---|
| Group 1 | Week 2 | Cloud forensics — acquiring and validating evidence from cloud services (IaaS/SaaS). |
| Group 2 | Week 2 | Forensic implications of full-disk and application-level encryption. |
| Group 3 | Week 3 | SSD forensics — how TRIM and wear-levelling affect data recoverability. |
| Group 4 | Week 3 | Recent advances in file carving and fragmented-file recovery. |
| Group 5 | Week 4 | Reconstructing user activity from Windows artefacts with current open-source tooling. |
| Group 6 | Week 4 | Detecting anti-forensics — timestamp manipulation, wiping and artefact tampering. |
| Group 7 | Week 5 | Memory forensics for detecting fileless and in-memory malware. |
| Group 8 | Week 5 | Machine-learning approaches to digital-forensic triage and malware detection. |
| Group 9 | Week 6 | Network forensics of encrypted traffic (TLS 1.3, QUIC, DNS-over-HTTPS). |

| Group | Week | Topic |
|---|---|---|
| Group 10 | Week 6 | Investigating phishing and online financial fraud — with a Pakistan focus. |
| Group 11 | Week 7 | Mobile forensics on modern, hardened Android and iOS devices. |
| Group 12 | Week 7 | IoT and smart-device forensics — sources of evidence and extraction methods. |

*Each group's weekly presentation, report and research feed the shared project — the Lab & SOP Handbook and the running mock case.*